



HYPERLOOP IN REVIEW

16 July 2024

Hyperloop's evolving landscape
from January to June 2024

hyperloop
development
program



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PREFACE

Dear hyperloop enthusiast,

I am very pleased to be part of an exciting ecosystem since the beginning of this year. I would like to take this opportunity to look back at the past six months. A period – no doubt – that many of us will remember as a most dynamic time for the hyperloop movement.

As we are editing this half-year report we are doing so to publish, release and discuss the developments during the European Hyperloop Week. An event which will take place in Switzerland this year. A country with a unique railway culture and a track-record of innovation with care for precision and safety.

This ‘Hyperloop in Review’ report is published by the Hyperloop Development Program. We will share our observations with you - which probably are by no means complete - without giving much interpretation. Nonetheless, I want to use this foreword to share you my reflections on all the industry dynamics of the past half year and share you my personal perspectives here and now.

The first half of 2024 was marked - in negative ways - by the events surrounding HyperloopOne closing down and even more so by the media attention surrounding this event in the aftermath. Additionally, the other US-based hyperloop promoter HyperloopTT ended up in financial uncertainty further fueling the negative sentiment surrounding hyperloop that was already gaining media attention.

The first half of 2024 was - however - also marked by many great successes and by pre-conditions and industry enablers being delivered in the European ecosystem:

- At the EU level the hyperloop has been adopted as an opportunity in the TEN-T regulation and the European Commission’s impact assessment for regulatory and legislative opportunities is ongoing.
- Europe’s Rail Joint Undertaking (EU-Rail) launched - and is currently evaluating proposals for - a call to develop a blueprint for hyperloop deployment in Europe based on a standardized concept.
- Strategic infrastructures for research and development are being established with DemoTube in Switzerland, goTube in Germany, and the European Hyperloop Center in the Netherlands.
- The first ever European tender for a hyperloop route - the Hyper Transfer project - has been completed and awarded in Italy to a consortium led by Webuild and Leonardo.

All of the above brings us at the HDP to the view that 2024 is a year in which hyperloop has its reality check. A year in which it is no longer about the dream-value of hyperloop, but about delivering on hardcore product development, certification and industrialization in line with the reality and expectation of a transportation industry that carries the most precious of cargo: you, me, our families and friends.

Having embarked this journey, I would like to extend our heartfelt thank you to the HDP Partners for their invaluable collaboration and continued support for bringing forward hyperloop development.

Sincerely,

Klaus Rudischhauser



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1. TOWARDS A NEW TRANSPORT TECHNOLOGY

Hyperloop is being developed within the wide realm of high-speed guided transport and can significantly improve connections across regions and countries. Notably, being energy efficient during operations, the new transport mode will contribute to reducing energy consumption in transport and mobility on long distances.

Reinventing existing technologies

The hyperloop combines existing technologies of magnetic levitation trains and a low-pressure environment in tubes. A hyperloop vehicle can therefore move without rolling resistance and with very limited air resistance. The tubes have a low-pressure environment maintained by air pumps. This enables movement with minimal aerodynamic resistance and low energy consumption. The system can be scaled up for high-frequency transport, which is comparable to the efficiency of today's metro or light rail connections.

Enabling hybrid transportation

Hyperloop is designed for passengers and cargo. Vehicles, resembling small airplanes without wings, move through a tube. The vehicles use magnetic levitation, guidance, and propulsion, which consists of steel rails in the tube and magnets on the vehicles. Safe and efficient access for people and goods to the vehicle is being designed through applying airlocks.

Enhancing multimodal solutions

Hyperloop transport requires seamless integration within a multimodal network to reach its full

potential. Transport hubs will facilitate the boarding of passengers or the transfer of cargo, as well as the transit to intercontinental aviation at airports and first- and last mile solutions within distribution centers.

Strengthening continental & regional transport

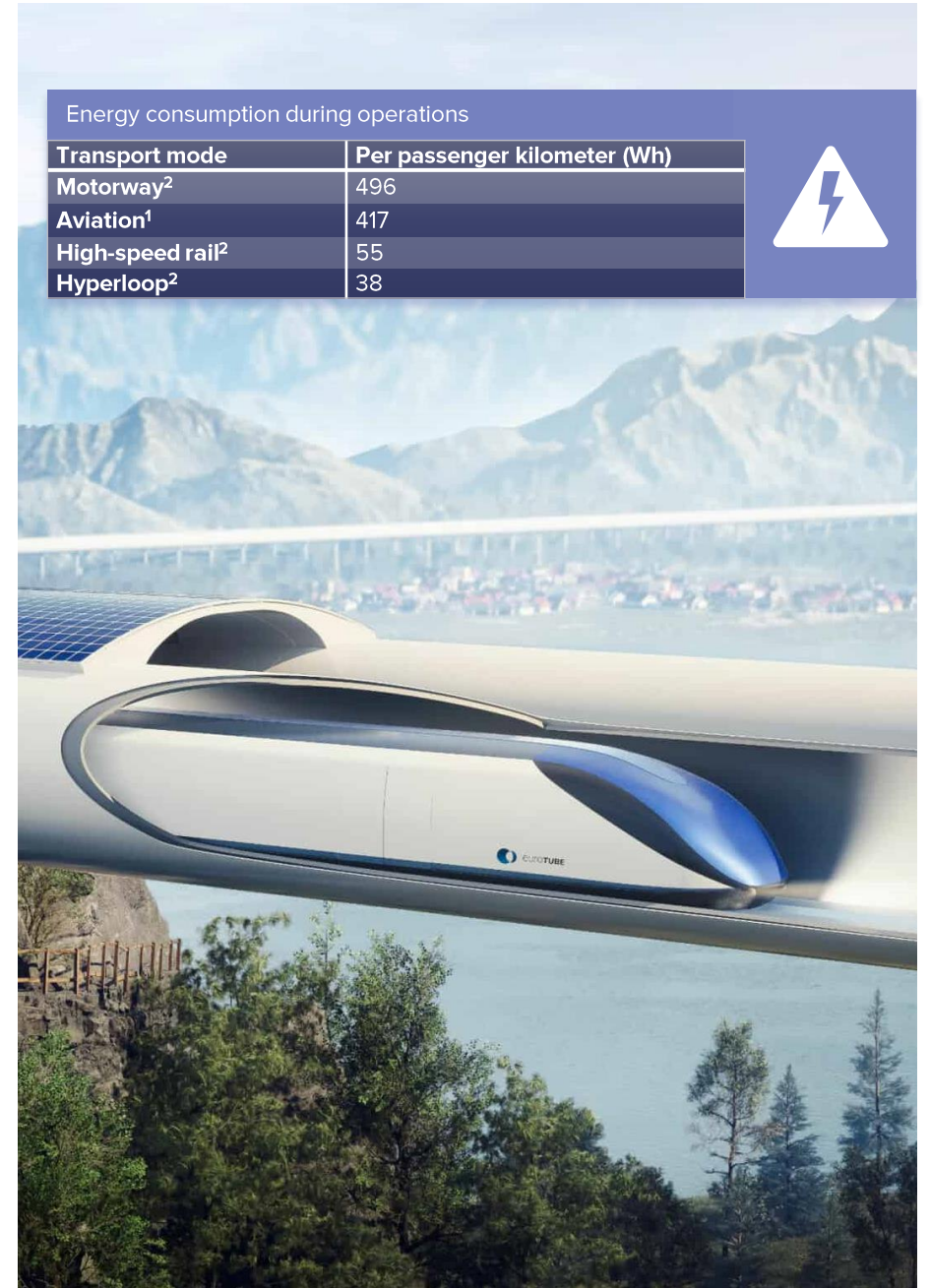
Within a European network the cruising speed is expected to be 700 km/h, according to the HDP's Hyperconnected Europe report (2022). Therefore, the hyperloop has the potential to alleviate transport over long distance, such as the strain on busy airports and congested motorways or railway networks. For different sorts of distances, the speed can be easily adjusted, making the hyperloop a versatile transportation solution for connections across a continent or in densely populated metropolitan regions.

Lowering costs with the hyperloop

The hyperloop infrastructure's construction costs are comparable to high-speed rail due to its modular structure and relatively small spatial footprint, based on a CAPEX comparison executed by Deutsche Bahn E.C.O. Group and Hardt Hyperloop in 2022. In addition, the hyperloop's efficiency is increased by the friction-less magnetic propulsion and levitation as well as an enclosed environment, protecting the moving vehicles from environmental conditions. These features result in less wear and tear and, as a result, lower maintenance costs.

Energy consumption during operations

Transport mode	Per passenger kilometer (Wh)
Motorway ²	496
Aviation ¹	417
High-speed rail ²	55
Hyperloop ²	38



1. Oak Ridge National Laboratory, Transportation Energy Data Book, 2022

2. Antea Group and Hardt Hyperloop, Factsheet 'Waarom de hyperloop vanuit het stikstofvraagstuk zo interessant is', 2023

Construction of DemoTube test facility in Dübendorf, Switzerland





2. LATEST DEVELOPMENTS

Below is a chronological overview of events from January until June 2024.

January

- [Webuild, Leonardo and partners](#) win the European tender for a hyperloop route – the Hyper Transfer project – in Veneto, Italy.
- [Europe's Rail Joint Undertaking](#) (EU-Rail) launches a call for proposals, which is aimed at the technical harmonization, the design concept, and roadmaps on industrialization for the hyperloop.
- [ArcelorMittal](#) pairs up with Indian partners to develop hyperloop.

February

- The [China Aerospace Science and Industry Corporation](#) successfully performs a test with hyperloop-related technology, breaking a speed record of more than 600 km/h on a 2 km test track in Shanxi, China.
- The European Hyperloop Center and the EuroTube Foundation decide to closely work together within a [European hyperloop infrastructure cluster](#) that is facilitated by the Hyperloop Development Program.
- [Klaus Rudischhauser](#), former Ambassador of the EU to Mexico, joins the HDP as Director General.

March

- The [European Hyperloop Center](#) draws global news coverage after unveiling its infrastructure that is fully equipped for testing and demonstrating.
- [Indian Railways](#) launches a tender for a demonstrative hyperloop project.

April

- The Institute of Hyperloop Technology starts the construction of the [goTube hyperloop facility](#) for which they acquired magnetic propulsion technology and power electronics from Polish hyperloop developer Nevomo.
- [Zeleros](#), the Spanish hyperloop developer, attracts new investment and embarks on a new stage by focusing on the development of batteries, electrification technologies, and transportation automation systems.
- The [Indian Institute of Technology](#) in Madras announces a hyperloop competition for 2025.



May

- In [China](#) concepts are being developed for a very first hyperloop corridor by 2035, connecting Shanghai with Hangzhou over a distance of 150 km.
- [EuroTube](#) starts the assembly of its hyperloop test facility, spanning 120 meters, in Dübendorf, Switzerland.
- [Swisspod](#) finishes the first construction phase of their hyperloop test track in Colorado, USA.

June

- The [EU](#) recognizes hyperloop as an opportunity for more sustainable technology in its transport network policy (TEN-T).
- The [German Railway Authority \(Eisenbahn-Bundesamt\)](#) initiates a research project for assessing the current state of hyperloop development. The project aims to provide objective results, including information gaps.
- Student team [Delft Hyperloop](#) demonstrates their lane switch technology at the Technological University of Delft.

Hyperloop test track at the Indian Institute of Technology Madras in Chennai, India





3. HYPERLOOP IN POLICY MAKING

Hyperloop has been put forward in policy making, strategic plans, standardization and certification procedures, position papers and recently published studies.

TEN-T

The revised [Trans-European Transport Network \(TEN-T\) regulation](#) has been adopted in June 2024. The regulation now stresses that the European Commission and EU Member States should encourage projects, which aim to promote and deploy sustainable emerging technologies that enhance and ease the transport and mobility of passengers and freight. One of the examples to encourage the realization of more sustainable transportation is to include new railway technologies such as hyperloop. This recognition of hyperloop as a key technology and future transport mode highlights its potential to bring impactful added value through more sustainable, efficient, and high-speed travel options for both passengers and freight.

Europe's Rail

In its [work plan for 2024](#), Europe's Rail identifies new approaches for guided transport modes, including magnetic levitation and vacuum tube technique, that reduce energy consumption, noise, and pollutant emissions as "an important and possibly unavoidable component of the mobility of the future."

EU regulatory framework

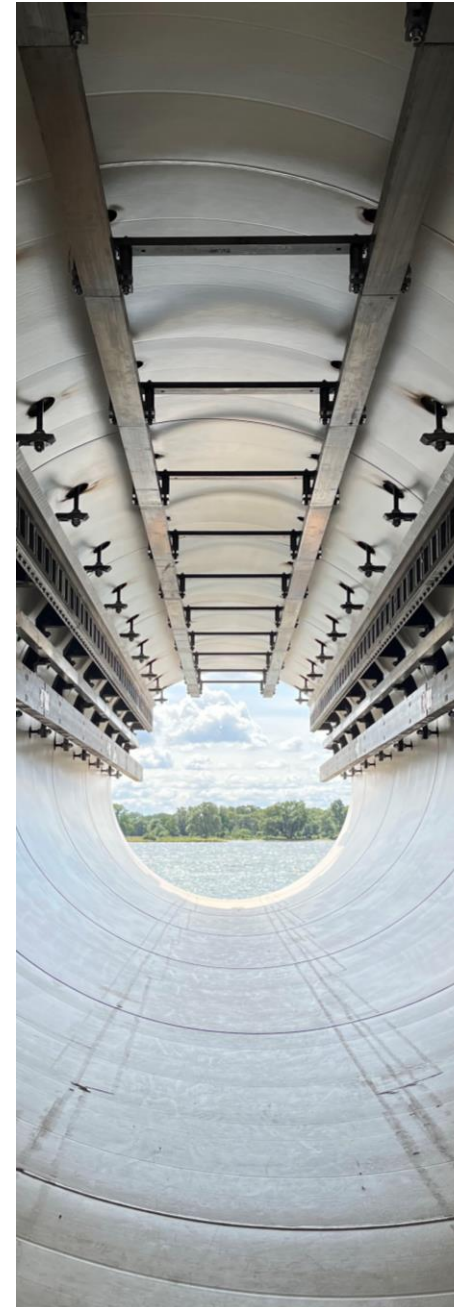
The Hyperloop Development Program has positioned itself on the need for an [EU regulatory framework](#) for hyperloop. Legislation for the hyperloop should be introduced from the start and can spur technological convergence of hyperloop technology on the short term, whilst ensuring cross-border interoperability and high safety standards on the long term.

Certification for passenger transport

First concepts have been approved by TÜV Süd and the Technical University of Munich (TUM) for manned operation under a framework of [hyperloop guidelines](#) aligned with the railway industry development practices and common safety methods. TÜV SÜD specifically certified the first segment of TUM Hyperloop's test tube and passenger pod.

Benelux Union

Hyperloop has been adopted in the Benelux Union [annual plan for 2024](#). The Benelux Union and its Members States have initiated a study about the potential of hyperloop development and routes, which is expected to be published in the third quarter of 2024. More specifically, the annual plan mentions possible support for a cross-border pilot project.



Saudi Arabia

The [National Strategy for Transport and Logistics](#) of Saudi Arabia has set its objectives to be reached in the railway sector, moving forward towards advancing development in the Kingdom, and contributing to increasing the GDP. The Saudi railway sector is looking at many innovations in the field of train transport technology, including high-speed trains as well as hyperloop technology.

India

The [Ministry of Railways of India](#) aspires to have more than 5 hyperloop routes by 2047 in its long-term plan. The country follows a parallel approach: 1) develop demonstration projects with global leading players, and 2) develop an indigenous system. The overall objective is to demonstrate a full-scale pilot by 2026. A global tender is to be opened.

European Hyperloop Center test facility in Veendam, the Netherlands

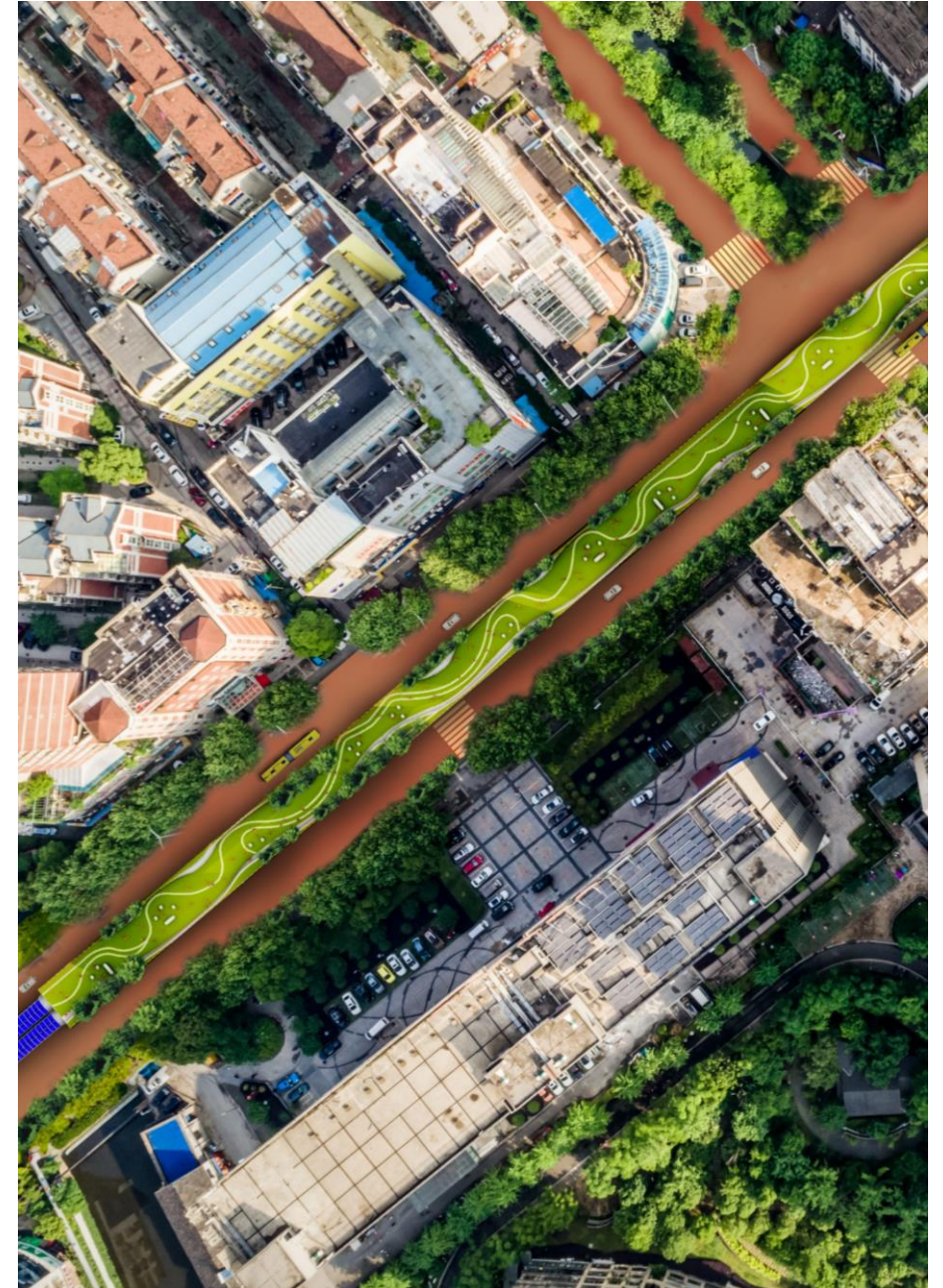




4. FUTURE OUTLOOK

For the second half of 2024 we see no loss of dynamics. Hyperloop – after all – is a fast modality:

- We expect an outcome of the **ERJU-FA7 project** by the end of the summer and a start of this flagship project in the fourth quarter of 2024.
- The **European Hyperloop Week** will take place in July, during which university teams present prototype components and pods to a panel of judges, competing in several categories. Moreover, the first phase of realizing EuroTube's test facility **DemoTube** will be inaugurated during this week.
- We hope to see a positive outcome of **the EU-level hyperloop impact assessment** for taking regulatory measures. We are keen on learning what the European Commission will propose for preparing next steps in the European legislative process.
- In September, **TEKNOFEST**, Türkiye's aerospace and technology festival, will include a hyperloop competition for magnetic levitation and suspension technologies.
- We are eager to see **impressive new test facilities** coming to life, amongst which:
 - The European Hyperloop Center (EHC) in Veendam, the Netherlands. In July, Hardt Hyperloop already performed a successful vacuum test at the EHC. In September, the EHC Grand Event will be held with a showcase of hyperloop technology.
 - The Institute of Hyperloop Technology's goTube test infrastructure in Emden, Germany, where Nevomo's magnetic propulsion technology and power electronics will be applied. The tube has already been installed and interior construction is underway.
 - Testing by TuTr Hyperloop at the hyperloop test track located at the Indian Institute of Technology Madras in Chennai, India.
- We look forward to the completion of the first part of the **Hyper Transfer study** in Italy, aimed at connecting the cities of Venice and Padua with the hyperloop.





goTube test facility in Emden, Germany



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